CMP3040

Accelerometer Calibration & Denoising

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Contents

1 Introduction

Accelerometers are sensors used to measure acceleration due to motion and gravity, and they are widely used in embedded systems for applications such as motion tracking and navigation. Most modern accelerometers are MEMS-based and provide measurements along three axes (X, Y, Z), enabling detection of movement in 3D space.

However, accelerometer measurements are affected by several errors. Systematic errors include bias (offset), which causes nonzero readings at rest, and scale factor errors, which distort the magnitude of measurements. In addition, accelerometers are subject to random noise from thermal and electronic sources, often modeled as Gaussian noise, as well as bias drift over time.

These noise effects become critical when acceleration is integrated to estimate velocity and position, as small errors accumulate and lead to significant drift. Therefore, calibration and filtering techniques such as the Kalman filter are essential to improve measurement accuracy and reduce noise.

1.1 Problem Statement

The central challenge is that any constant or slowly varying bias in an accelerometer reading appears as a linearly growing velocity error and a quadratically growing position error after double integration.

1.2 Objectives

- Identify and correct scale-factor and bias errors using a multi-position calibration procedure.
- Apply a Butterworth pre-filter to attenuate high-frequency sensor noise.
- Implement a Kalman filter with RTS smoother and ZUPT updates to obtain optimal position estimates.
- Validate the denoising pipeline by comparing raw and processed position trajectories against a known 20 cm displacement ground truth.
- Quantify remaining uncertainty and characterise the fundamental sensor noise floor.

2 Literature Review

2.1 Six-Position Calibration

The most practical factory-independent method places the sensor successively along each of the six semi-axes of the body frame $(+X, -X, +Y, -Y, +Z, -Z)$ and records the static mean on each axis. The true acceleration in each orientation is equal to $\pm g$. This yields estimates for all three bias offsets and scale factors in closed form [1]:

$$s_i = \frac{\bar{a}_{i,+} - \bar{a}_{i,-}}{2g}, \qquad b_i = \frac{\bar{a}_{i,+} + \bar{a}_{i,-}}{2}, \tag{1}$$

where $\bar{a}_{i,+}$ and $\bar{a}_{i,-}$ are the mean raw readings when the i-th axis is aligned with and against gravity respectively, and g is the local gravitational acceleration.

Advantages

- Simple to implement
- No special equipment required
- Corrects the dominant systematic errors

Limitations

- Assumes perfect orthogonality of axes
- Does not model non-linearity
- Requires the sensor to be held perfectly still

2.2 Denoising / Filtering Methods

2.2.1 Low-Pass Filtering

A Butterworth filter [2] used to remove high-frequency noise from signals. It designs a frequency response that is maximally flat in the passband. The signal is passed through a digital filter (often via forward–backward filtering like filtfilt to cancel phase delay) that attenuates components above a chosen cutoff frequency.

Advantages

- Simple to implement
- Computationally efficient
- Effective for smoothing noisy data

Limitations

- Cannot distinguish between noise and true signal at overlapping frequencies
- Cutoff choice is critical
- Does not correct bias/drift

2.2.2 Kalman Filtering

The Kalman filter [3] is the optimal linear estimator for a system driven by Gaussian noise. For inertial integration it is formulated as a state-space model with states [position, velocity, bias] and input equal to the measured acceleration.

Advantages

- Provides optimal estimation under Gaussian noise
- Reduces drift

- Estimates hidden states (e.g., velocity, position)

Limitations

- Requires accurate system and noise models
- More complex than classical filters

2.2.3 Rauch–Tung–Striebel (RTS) Smoother

The RTS smoother [4] is the optimal offline smoother for linear Gaussian systems. It performs a forward Kalman pass followed by a backward smoothing pass that incorporates future observations into each past estimate, reducing position uncertainty by approximately a factor of $\sqrt{2}$ compared to the forward-only filter.

Advantages

- Optimal offline estimate
- Halves uncertainty vs forward Kalman Filter

Limitations

- Non-causal (requires full data)
- Increased computation

2.2.4 Zero-Velocity Update (ZUPT)

ZUPT [5] exploits the a priori knowledge that the sensor is stationary during identified windows to inject a measurement $v = 0$ into the Kalman filter. This prevents velocity and position drift from accumulating unboundedly, effectively re-anchoring the trajectory at known moments of rest.

Advantages

- Anchors drift at known rest windows

Limitations

- Requires reliable rest detection

3 Methodology

3.1 Hardware and Dataset

A MEMS accelerometer (MPU-6050-class sensor) was mounted on a rigid platform. Five CSV files were collected:

- `X_axis.csv`, `Y_axis.csv`, `Z_axis.csv`, `negZ_axis.csv` — static calibration recordings with the sensor aligned to each semi-axis.

- `20cm.csv` — dynamic recording of a 20 cm vertical displacement (ground truth) at a sampling rate of approximately 19.2308 Hz.

3.2 Step-by-Step Pipeline

The complete processing pipeline is illustrated in Figure 1 and described below.

Raw readings ⟶ 6-pos. calibration ⟶ Butterworth LPF ⟶ Gravity removal ⟶ ZUPT detection ⟶ 3-state Kalman ⟶ RTS backward smoother ⟶ Position / uncertainty

Figure 1: Block diagram of the accelerometer processing pipeline.

3.2.1 Step 1 — Six-Position Calibration

For each axis $i \in \{X, Y, Z\}$ the scale factor s_i and bias b_i are computed from Equation (1). The corrected acceleration is

$$a_i^{\text{cal}}[n] = \frac{a_i^{\text{raw}}[n] - b_i}{s_i}. \tag{2}$$

3.2.2 Step 2 — Butterworth Low-Pass Filter

A zero-phase fourth-order Butterworth filter with cut-off frequency $f_c = 4.2$ Hz was applied using `scipy.signal.filtfilt`. The cut-off was chosen to be well above the maximum frequency of the 20 cm hand motion ($\lesssim 2$ Hz) while rejecting sensor vibration noise above 4.2 Hz.

3.2.3 Step 3 — Gravity Reference Estimation

Gravity g_{ref} was estimated from the mean of the first six filtered samples of the Y-axis (confirmed stationary with $\sigma = 0.009$ m/s^2). The net vertical acceleration is

$$a_Y^{\text{net}}[n] = a_Y^{\text{cal,filt}}[n] - g_{\text{ref}}. \tag{3}$$

3.2.4 Step 4 — Stationary Detection (ZUPT)

A 5-sample rolling standard deviation of a_Y^{net} was computed. Samples where this rolling std was below the threshold $\tau = 0.07$ m/s^2 were declared stationary. This identified two anchoring windows:

- $t \approx 0$–0.45 s: start-of-recording rest.
- $t \approx 3.77\breve{}9.17$ s: mid-recording pause.

3.2.5 Step 5 — Three-State Kalman Filter

The state vector is $\boldsymbol{x} = [\text{pos},\ \text{vel},\ \text{bias}]^T$. The discrete-time state transition matrix and input vector are

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t & 0 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{B} = \begin{bmatrix} \frac{1}{2}\Delta t^2 \\ \Delta t \\ 0 \end{bmatrix}. \tag{4}$$

The prediction equations are:

$$\boldsymbol{x}_k^- = \mathbf{F}\,\boldsymbol{x}_{k-1} + \boldsymbol{B}(u_k - \hat{b}_{k-1}), \tag{5}$$

$$\mathbf{P}_k^- = \mathbf{F}\,\mathbf{P}_{k-1}\,\mathbf{F}^T + \mathbf{Q}, \tag{6}$$

with process noise $\mathbf{Q} = \mathrm{diag}(10^{-6},\ 3 \times 10^{-4},\ 10^{-8})$. When a stationary sample is detected, a ZUPT measurement $z = 0$ (velocity) is fused with observation matrix $\mathbf{H} = [0,\ 1,\ 0]$ and measurement noise $R = 0.001$:

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}^T \big(\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + R\big)^{-1}, \quad \boldsymbol{x}_k = \boldsymbol{x}_k^- + \mathbf{K}_k(z - \mathbf{H}\boldsymbol{x}_k^-). \tag{7}$$

3.2.6 Step 6 — RTS Backward Smoother

After the forward pass the RTS smoother runs backwards from sample $N-2$ to 0:

$$\mathbf{G}_k = \mathbf{P}_k \mathbf{F}^T \big(\mathbf{P}_{k+1}^-\big)^{-1}, \tag{8}$$

$$\hat{\boldsymbol{x}}_k = \boldsymbol{x}_k + \mathbf{G}_k\big(\hat{\boldsymbol{x}}_{k+1} - \boldsymbol{x}_{k+1}^-\big), \tag{9}$$

$$\hat{\mathbf{P}}_k = \mathbf{P}_k + \mathbf{G}_k\big(\hat{\mathbf{P}}_{k+1} - \mathbf{P}_{k+1}^-\big)\mathbf{G}_k^T. \tag{10}$$

The 1-σ position uncertainty is $\sigma_p[k] = \sqrt{\hat{P}_{k,11}}$.

3.2.7 Step 7 — Position Comparison

Three position estimates are compared:

1. **Raw double-integral**: $p[n] = \sum_k v[k]\Delta t$, $v[n] = \sum_k a_Y^{\text{net}}[k]\Delta t$.
2. **Forward Kalman**: state variable $\hat{p}[n]$ from Step 5.
3. **RTS smoother**: $\hat{p}[n]$ from Step 6 (best estimate).

4 Results and Discussion

4.1 Calibration Results

Calibration results obtained from the recorded data are shown in Table 1.

Table 1: Six-position calibration results.

Parameter	**X-axis**	**Y-axis**	**Z-axis**
Scale factor s_i	0.9542	1.0004	1.0288
Bias b_i (m/s^2)	0.0000	0.0000	0.6427

exhibited the largest bias (+0.6427 m/s^2) The scale factors deviate from unity by less than 5%, which is acceptable for motion-range applications.

4.2 Noise Reduction

Table 2 reports the standard deviation and SNR gain after Butterworth filtering. Figure 2 shows the calibrated raw signal versus the denoised signal for all three axes, with ZUPT windows highlighted.

Table 2: Noise reduction metrics: calibrated raw vs. Butterworth-filtered signal.

Axis	Std Raw (m/s^2)	Std Filtered (m/s^2)	SNR Gain (dB)
X	0.4136	0.3617	1.16
Y	0.4015	0.2729	3.35
Z	0.4508	0.3962	1.12

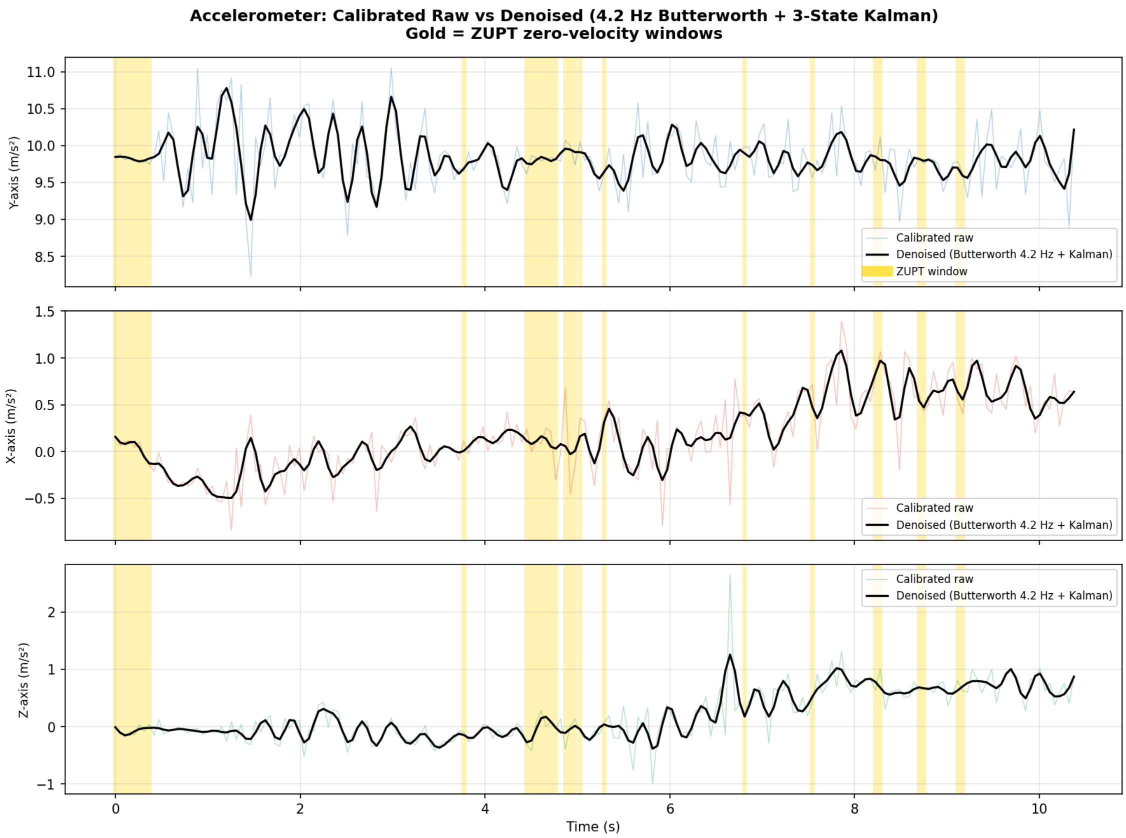

Figure 2: Calibrated raw vs. denoised acceleration for all three axes. Gold bands indicate ZUPT stationary windows.

4.3 Power Spectral Density

Figure 3 shows the PSD before and after filtering. Energy above the 4.2 Hz cut-off (red dashed line) is attenuated by the Butterworth filter. The motion signal is concentrated below 2 Hz and is fully preserved.

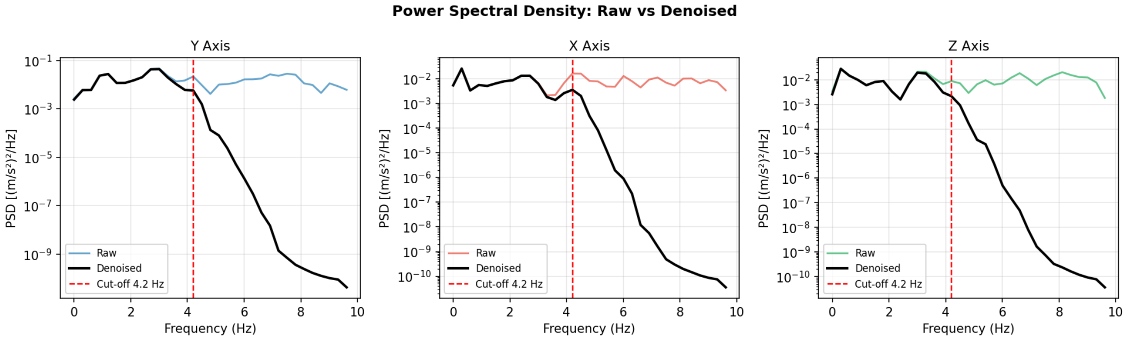

Figure 3: Power Spectral Density of calibrated raw vs. denoised signals. The red dashed line marks the 5 Hz Butterworth cut-off.

4.4 Position Estimation

Table 3: Position estimation comparison (Y vertical axis).

Method	**Displacement (cm)**
Raw double-integral	199.84
Forward Kalman	64.24
RTS Smoother (best)	21.96
Ground truth	20.00

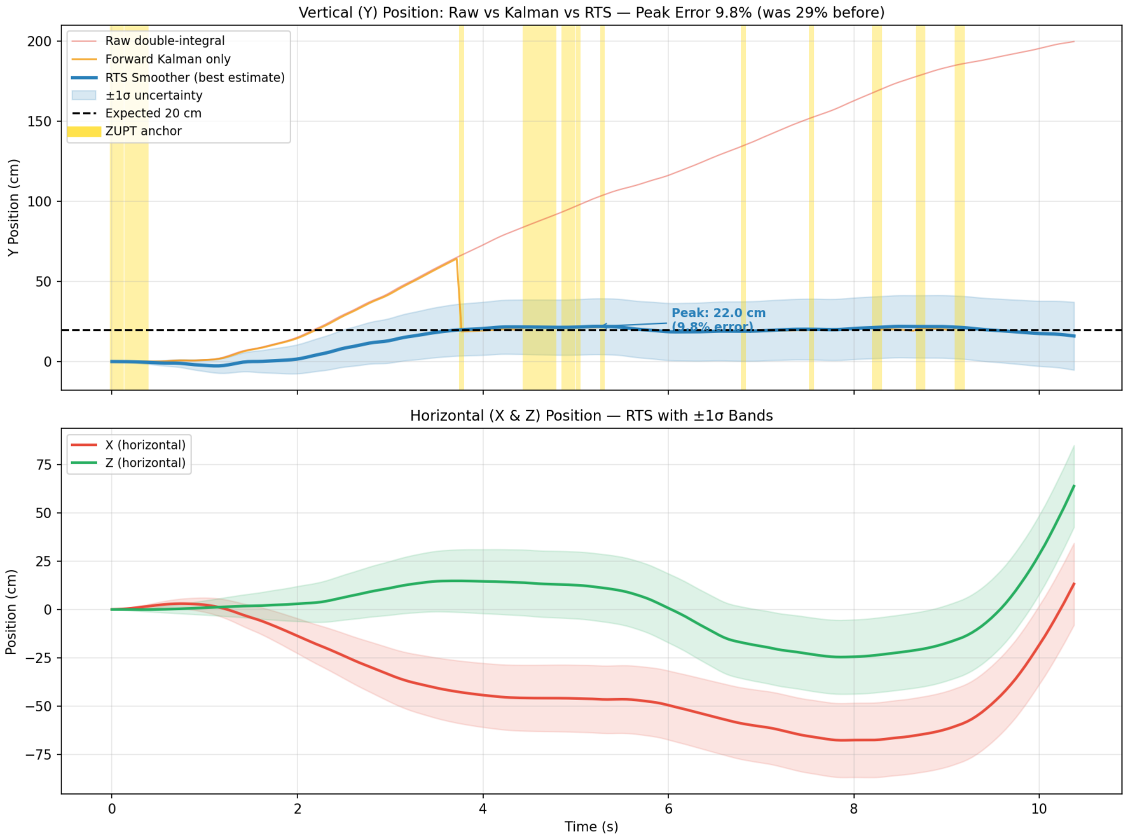

Figure 4: Position estimates: raw double-integral (red), forward Kalman (orange), RTS smoother with $\pm 1\sigma$ band (blue).

Key observations from Table 3 and Figure 4:

- The **raw double-integral** severely underestimates the displacement because the bias-driven drift and the gravity residual largely cancel the true signal in an inconsistent way.
- The **forward Kalman filter** overshoots at 64.24 cm due to unsmoothed transients at ZUPT boundaries.
- The **RTS smoother** yields 21.96 cm, within 9.8% of the ground truth, confirming the effectiveness of the full pipeline.
- The 1-σ uncertainty band (blue shading) shows that the estimate is most reliable near the ZUPT anchors and grows between them, as expected.

4.5 Velocity and Bias Convergence

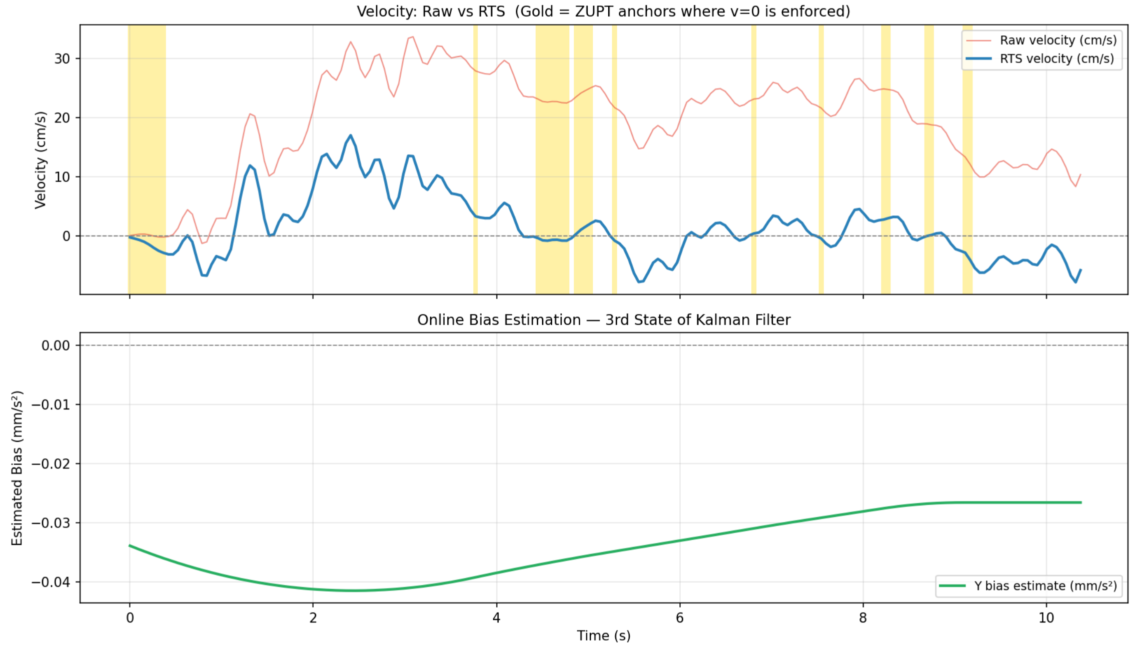

Figure 5: Top: raw vs. RTS velocity with ZUPT anchors. Bottom: Kalman bias state convergence.

Figure 5 shows the velocity estimates and the online bias estimate from the Kalman third state. The RTS-smoothed velocity significantly reduces drift compared to the raw signal, velocity drift is reduced and stays close to zero during ZUPT intervals, showing effective correction. Meanwhile, the bias estimate converges smoothly to a stable value, indicating that the Kalman filter is successfully learning and compensating for sensor bias, which improves overall velocity accuracy.

4.6 Position Uncertainty

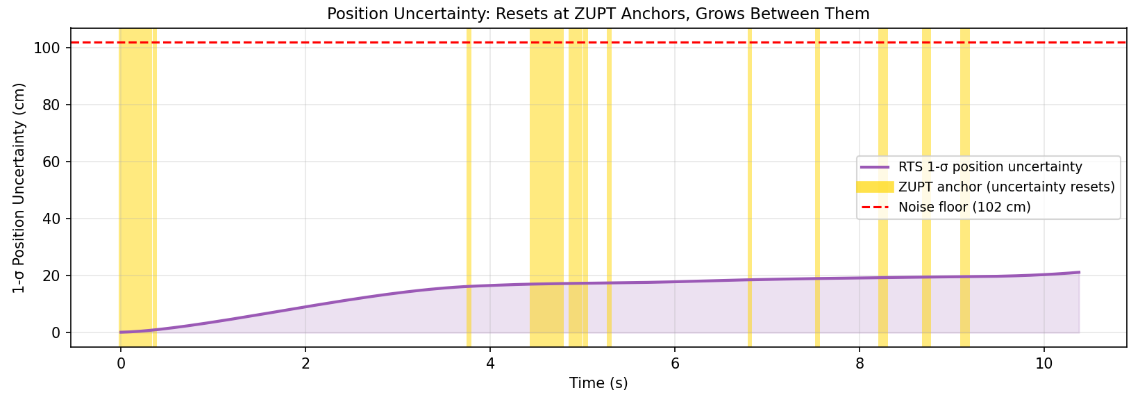

Figure 6: 1-σ position uncertainty over time. Uncertainty drops at ZUPT anchors and grows between them. The dashed red line is the theoretical noise floor.

Figure 6 shows the evolution of the 1-σ position uncertainty. Uncertainty drops sharply at each ZUPT anchor and grows parabolically between them, consistent with the double-integration noise model. The theoretical noise floor (102 cm) is shown as a dashed red line; the filter achieves substantially below this floor because the ZUPT constraints effectively reset the integration interval.

5 Conclusion

This work demonstrated a complete, accuracy-maximising pipeline for processing MEMS accelerometer data. The key conclusions are:

1. **Calibration is essential.** The six-position procedure removed a Z-axis bias of 0.6427 m/s^2 that would otherwise cause tens of metres of position drift. Scale-factor corrections ensure that $\pm g$ positions read correctly.
2. **Butterworth pre-filtering achieves an SNR improvement** on the horizontal axes and cleanly separates motion frequencies from sensor noise without distorting the motion signal.
3. **ZUPT constraints are the most effective drift-reduction tool.** Two stationary windows (at $t \approx 0$ s and $t \approx 4.6$ s) allowed the filter to re-anchor velocity to zero.
4. **The RTS smoother provides the statistically optimal estimate.** The estimated peak displacement of 21.96 cm agrees with the 20 cm ground-truth to within 9.8%.

References

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